

shortages are frequent. This can create a mess in the healthcare supply chain. It forces providers to either purchase alternatives that are much more expensive or maintain a comprehensive backup inventory of products that are at risk of being in short supply. This then leads to the added cost of inventory management and product expiration. The best option for dealing with this is the use of technology that assists with early notification and response, especially in the case of certain surgical agents, cardiovascular agents and IV fluids. The technology enables providers to more effectively manage and monitor the long-standing shortages of hundreds of drugs, and to be prepared for unforeseen shortages in other drugs.

Integration issues: Healthcare practices, facilities and hospitals are becoming more consolidated. Health systems are growing, merging and acquiring. As this happens, supply chains within these siloed yet merged organisations remain separate. Healthcare providers need to address

this because inconsistency between these supply chains will negatively impact the bottomline. These supply chains need to be integrated. Purchasing channels need to be centralised.

Facilities need to share contracts so that they can gain access to higher tier pricing. Without these adjustments, the supply chain will not be cost-effective and the processes that it contains will remain inefficient.

Variation in preferences:

Different physicians and healthcare professionals have different preferences for the products that they use. Most organisations allow this to be the final say in purchases. Unfortunately, physician preferences are rarely tied to more successful outcomes and can often lead to cost variation. The fact is not that physicians do not care about the higher cost or that the product does not offer better patient outcomes. Rather, it is that they are unaware of the true cost of their choices. A study found that surgeons correctly estimate the cost of items just

over one-fifth of the time. Helping physicians to understand the costs and the alternatives can help them find products that will offer comparable or better outcomes at a better price.

Poor workflow design: Throughout the healthcare supply chain, many processes are unnecessarily duplicated. The reason for this is that the systems and entities that make up the supply chain are disconnected. Many of the tasks can be automated and integrated in a way that will allow all participants to share information more freely. The healthcare supply chain is complex and delicate. So, while these challenges can and should be addressed, it will take the right technology and solutions to do so.

Architecture of blockchain solution for the medical supply chain

Implementing a medical supply chain solution involves handling people (governance), processes (activities and information flow) and technology (using blockchain smart contracts

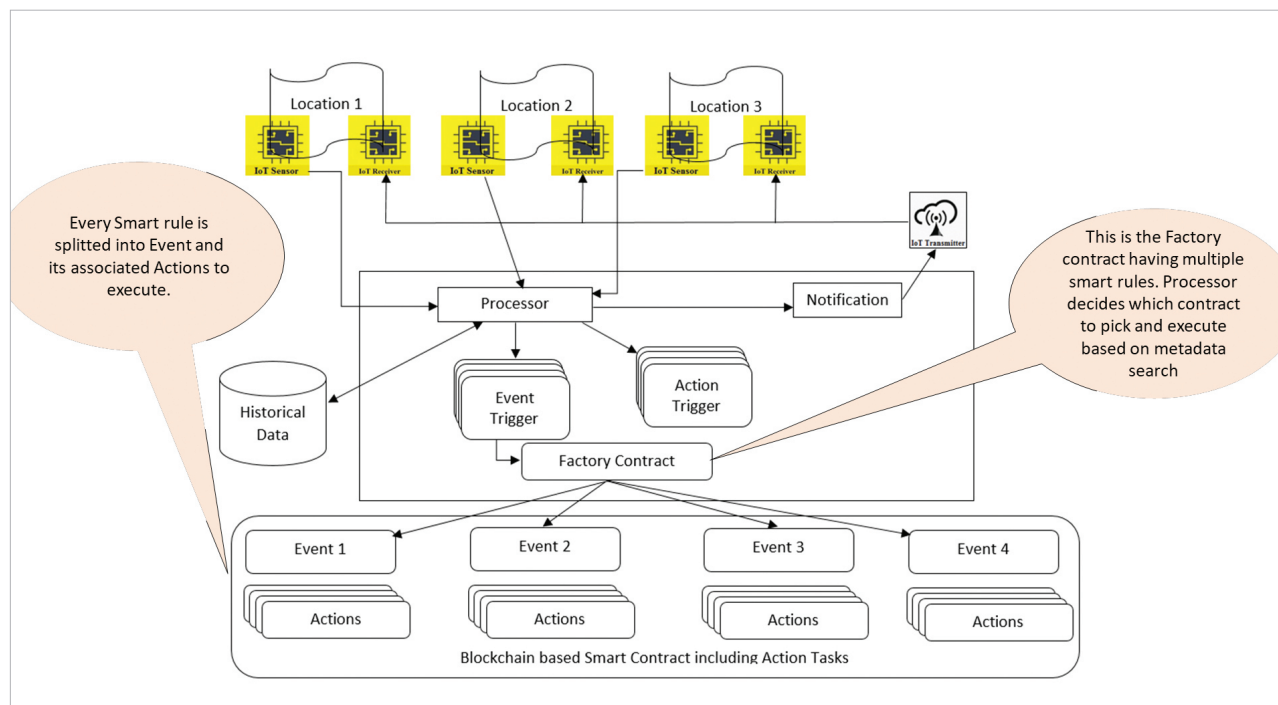


Figure 4: Blockchain based electronic medical record sharing between hospitals